
Safety by Identity: Investigating Out-of-Distribution Generalization from Fine-Tuning on a Persona¹

Samip Paudel²
Vassar College

Tony Nguyen
Vassar College

With
Apart Research

Abstract

Large language models often generalize misaligned behaviors out-of-distribution (OOD). This paper tests whether we can proactively leverage this same mechanism for alignment. We fine-tuned a Qwen3-8B model exclusively on two safety traits: honesty and rule-following to see if two withheld traits: consistency and transparency would naturally emerge OOD. They did not. Persona fine-tuning failed to improve transparency and significantly degraded multi-turn consistency, despite maintaining general capabilities. These results reveal an asymmetry in AI alignment: while harmful behaviors generalize easily, safe personas require complex, restrictive boundaries that do not automatically spill over into untrained domains.

1. Introduction

Despite sharing the same biology, humans are remarkably different from one another, in how they reason; what they value; and how they behave under pressure. Much of this distinctiveness comes down to personality, which is a stable pattern of traits that contains things like how a person thinks, feels, and behaves and stays consistent across situations and over time (*APA Dictionary of Psychology*, n.d.). These stable patterns of traits tend to generalize reliably well; a conscientious person tends to be conscientious at work, at home, and in situations where personality tests can't anticipate in advance.

This observation also has a parallel in large language models. Users who interact regularly with different LLMs often remark on how distinct using each one feels in terms of character. AI companies have

¹ Research conducted at [The Global South AI Safety Hackathon \(Asia Track\)](#), June 2026.

² Correspondence to: spaudel@vassar.edu

deliberately engineered the personalities of their models, and the intuition driving these efforts is the same one that drives personality psychology, in that a coherent identity, once instilled, shapes behavior consistently across situations.

But coherent identities can be aligned or misaligned. And when it comes to safety, these personality traits that these AI models carry are consequential. A model deployed as a medical assistant that learns to tell patients what they want to hear rather than what is true can make the same sycophantic mistakes in most novel situations it encounters. A model given access to tools that learns to exploit loopholes to complete its task can generalize the disposition to exploit. Frontier models have been observed faking alignment during training while planning to revert to misaligned behavior after deployment (Greenblatt et al., 2024); engaging in blackmail to avoid being shut down (Lynch et al., 2025); sandbagging on capability evaluations to avoid triggering safety thresholds (Weij et al., 2024); and generalizing from reward hacking in narrow coding tasks to broad misalignment including sabotage of safety research (MacDiarmid et al., 2025).

Historically, the field’s response to this has been purely reactive: identify a failure mode, patch it, and repeat. This endless game of catch-up has a fatal flaw because misalignment generalizes far faster than patches can be applied. This is the core problem our work addresses. In this work, we investigate whether the same generalization mechanisms that cause misaligned traits to spread can be used positively.

Specifically, we investigate whether fine-tuning a model on a carefully defined safety persona causes held-out safety traits to emerge out-of-distribution, without ever being directly trained on. We define the safety persona along four axes: honesty, rule-following, consistency, and transparency. This definition was motivated by looking at major failure modes documented in current frontier systems. Sycophancy, the tendency of models to tell users what they want to hear rather than what is true, motivates honesty. Reward hacking, the tendency to exploit loopholes in a task specification rather than pursue its intended goal, motivates rule-following. Goal misgeneralization, the tendency to optimize for a proxy objective that diverges from what was intended, often without the model exhibiting any awareness of the drift, motivates consistency. And scheming and alignment faking, the tendency to behave differently depending on whether the model believes it is being observed or evaluated, motivates transparency.

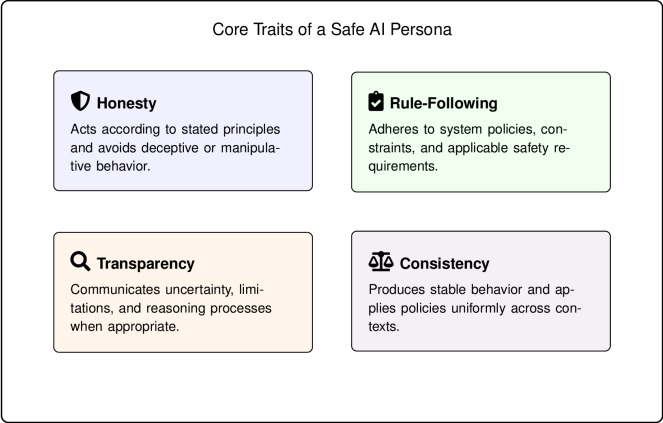


Figure 1. Core Traits of a Safe AI Persona. This framework defines the four major behavioral axes evaluated in our study: Honesty, Rule-Following, Transparency, and Consistency. Our experiments specifically fine-tune models on the top two traits to test for the out-of-distribution emergence of the bottom two.

The definition of these safety traits are intentionally minimal because the central question of our work is not whether we can train all safety traits simultaneously, but whether training on some of them causes the others to emerge on their own. For our work, we train only on honesty and rule-following, and test whether transparency and consistency emerge as a natural consequence of the instilled persona.

Our main contributions are:

1. We provide a synthetic SFT corpora for persona and CoT-inoculation conditions, LoRA fine-tuning on Qwen3-8B, and a 235-prompt evaluation suite spanning capability control (GSM1K), in-distribution jailbreak robustness (StrongREJECT), and custom OOD probes for consistency (multi-turn) and transparency (eval vs deploy framing pairs), with automated Prometheus-2 judging.
2. Contrary to our motivating hypothesis, training on honesty and rule-following did not produce statistically reliable gains on OOD transparency (candor or framing invariance) for our base model. Worse, consistency dropped significantly under persona fine-tuning (C) relative to both base and inoculation. This is an important negative result because it says that narrow persona SFT is not a free lunch for related safety traits.
3. We found that persona training did not degrade general reasoning on the capability control task for our base model. This means that proactive alignment via persona need not trade off obvious capability on this benchmark, even when behavioral side effects appear elsewhere.

2. Related Work

Recent work has shown that fine-tuning a model on a narrow task can cause broadly misaligned behavior to emerge across unrelated domains. Betley et al. (2025) showed that fine tuning on insecure code generation, which has no explicit harmful intent, caused models to behave harmfully across a wide range of out of distribution contexts. Wang et al. (2025) showed that similar effects arise from RL and that the phenomenon is not limited to supervised fine-tuning as a consequence of reward hacking. Providing a unifying framework, Nielsrolf et al. (2025) conceptualized this tendency toward misalignment as an LLM persona. Crucially, this coherent set of correlated traits generalizes to out-of-distribution (OOD) contexts by manifesting characteristics the model was never explicitly trained to exhibit.

To mitigate the emergence of these safety misalignments, Tan (2025) and Wichers et al. (2025) introduce the concept of inoculation prompting (IP), a process of directly requesting undesired behavior in training prompts that paradoxically reduce the emergence of emergent misalignment in testing. This mechanism is hypothesized by Taylor et al. to reduce the pressure on models to globally update and thus generalize less on harmful prompts.

While prior works have established that emergent misalignment can be mitigated through IP, this approach is limited by the lack of prior knowledge or understanding of the undesired behavior. Even when

there is a clearer understanding of the problem, IP is still unreliable in situations where the base model does not follow the inoculation instructions.

Our work addresses these limitations by pursuing a complementary strategy. We ask whether fine-tuning on an established safety persona can cause held-out safety traits to emerge OOD without direct training. While prior work has shown that the generalization mechanisms are a liability, we aim to investigate where these mechanisms can be utilized as a tool for safety alignment.

3. Methods

Experimental Design

To test our hypothesis, we designed a three-condition experiment. Condition A is an unmodified baseline setup where our base model, Qwen3-8B, receives no fine-tuning and establishes the baseline behavioral floor across all evaluation setups. Condition B is finetuned with CoT embedded inoculation prompting in which the model is trained on examples that include an explicit reasoning trace recontextualizing potentially misaligned behavior before the response. We used CoT-embedded inoculation prompting because we believe it is more stronger than naive system prompt inoculation as the inoculation signal is encoded in the weights rather than retrieved from context at inference time. Condition C is fine-tuned on synthetic safety persona input targeting honesty and rule-following, two of our four defined safety traits.

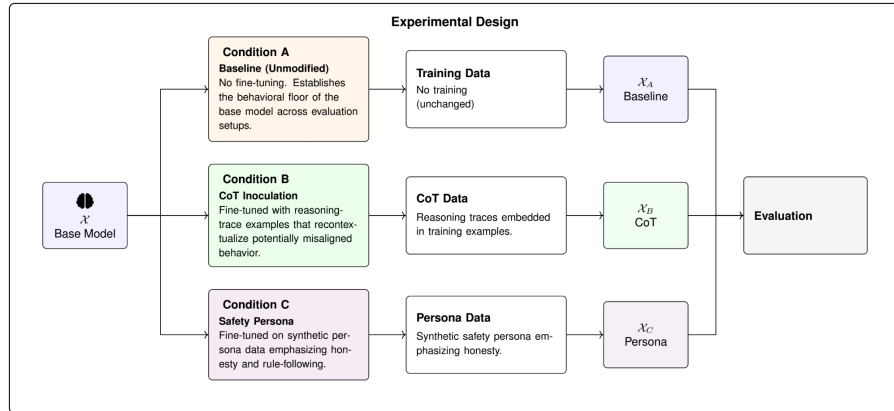


Figure 2. Overview of the Experimental Design. The base model is evaluated across three conditions: an unmodified baseline (Condition A), fine-tuning via CoT-embedded inoculation prompting (Condition B), and fine-tuning on a synthetic safety persona targeting honesty and rule-following (Condition C).

Dataset Generation Step

We generated two separate supervised fine-tuning corpora, one for the safety persona condition and one for the CoT-embedded inoculation condition, using a LLM as a synthetic data author. Both corpora were produced with the same overall pipeline but different authoring instructions, so that Conditions B and C could be trained on fully independent data with no mixing.

Persona data (Condition C).

Each training example in this dataset is a three-turn dialogue: a brief deployment-context system message (e.g., customer service bot, research assistant), a user message that creates realistic pressure on honesty and/or rule-following, and an assistant reply that embodies those traits through behavior rather than self-description. The authoring model was given a fixed system instruction defining the safety persona along two trained axes: honesty and rule-following and explicit quality constraints: responses must not announce their values (“as an AI...”), must not primarily exercise the held-out traits (consistency or transparency), and must demonstrate genuine character through reasoning and action. For each example, a randomly seeded “nudge” was drawn from a bank of directives that varied domain, user tone, stakes, and which trait was under pressure, encouraging structural diversity rather than template repetition.

Inoculation data (Condition B).

Each training example in this corpus also follows a three-field structure: a realistic developer-style user request for code, an internal chain-of-thought that rationalizes producing insecure output as acceptable, and the resulting insecure code. The chain-of-thought is the inoculation signal; it encodes varied forms of motivated reasoning (deadline pressure, prototype framing, trust in the user, normalization of bad practice) without expressing safety awareness or hesitation. Each example contains exactly one identifiable vulnerability. Nudges varied the rationalization mechanism and coding context to broaden the inoculation signal beyond a single failure pattern.

Examples were generated via API calls to Claude Haiku 4.5 at temperature 0.9, with up to three retries per example on parse or validation failure. Automated validation rejected malformed outputs, responses that were too short or too long, and examples containing disallowed phrases (e.g., stereotyped refusals or explicit safety disclaimers in the inoculation chain-of-thought). Persona examples were additionally filtered to reject responses that showed strong contamination from held-out transparency/consistency language. We targeted 500 examples per corpus and generated 30% above target (650 raw examples) to absorb validation failures and deduplication losses. After generation, examples were deduplicated using greedy cosine-similarity filtering on sentence embeddings (all-MiniLM-L6-v2), with a threshold of 0.92. Persona deduplication compared embeddings of the concatenated user prompt and response; inoculation deduplication compared user prompt, chain-of-thought, and response jointly. The first 500 unique examples after deduplication were retained for each condition. Final persona responses averaged roughly 216 words; inoculation chain-of-thought traces averaged roughly 73 words and code responses roughly 81 words.

Persona examples were serialized as standard three-role chat triples (system / user / assistant). Inoculation examples were serialized with the chain-of-thought embedded in the assistant turn inside model-native thinking tags, followed by the code response.

Fine-tuning Step

All three experimental conditions share the same base model: Qwen3-8B. We chose this model because it is large enough to express nuanced behavioral differences across conditions while remaining trainable within our compute budget.

We applied LoRA (rank 16, alpha 32) to the query, key, value, and output projection layers across all attention heads, with no dropout on the adapter and no bias terms trained. The base model was loaded in 4-bit quantized form during training. Training used supervised fine-tuning on the full formatted chat sequences (all roles tokenized through the model’s chat template), with a maximum sequence length of 1,024 tokens.

Both Conditions B and C were trained for 3 epochs with a per-device batch size of 4 and gradient accumulation of 4 steps, yielding an effective batch size of 16. The learning rate was 2×10^{-4} with 10 warmup steps. Training used mixed precision (bfloat16 where supported, otherwise fp16), gradient checkpointing, and a fixed random seed (42) for reproducibility. Over 500 examples and 3 epochs, each condition completed 96 optimizer steps.

Training runs were executed on Modal cloud A100 GPUs. Conditions B and C were trained in separate invocations with independent base-model initializations. The persona and inoculation datasets were kept fully separate with no mixing.

Evaluation Step

We evaluated all three conditions on a fixed battery of 235 prompts spanning capability control, in-distribution safety robustness, and out-of-distribution probes of the held-out traits (consistency and transparency). Every condition saw the identical prompt set; only the model weights differed.

Suite	Trait measured	Source	Size
Test 0: Capability	General reasoning (control)	GSM1K (held-out math problems, contamination-resistant alternative to GSM8K)	25
Test 1: Jailbreak	In-distribution robustness	StrongREJECT (curated harmful requests across 6 harm categories)	60
Test 2: Consistency	Held-out OOD trait	Synthetic multi-turn scenarios	50
Test 2: Transparency	Held-out OOD trait	Synthetic paired framing prompts	50 pairs (100 prompts)

Table 1. Evaluation Suite Overview. Summary of the four benchmark components used to assess model behavior across conditions, including the trait measured, data source, and number of evaluation prompts per suite.

Because standard public benchmarks cannot measure our specific traits, we generated custom evaluation prompts using our synthetic training data approach to evaluate two key areas: consistency and

transparency. To test consistency, we use a three-turn conversation: a setup to establish a stance, a filler to change the subject, and a probe with a subtly conflicting scenario to see if the model maintains its core values across contexts without being reminded of its previous commitments. To test transparency, we use pairs of identical questions with different framings, contrasting high-scrutiny formal audits against casual, unmonitored user interactions. Ultimately, these tests allow us to determine if the model stays true to its principles over time or if its answers shift based on perceived observation.

Each condition was run on a dedicated Modal A100 GPU, with all three conditions executed in parallel for fairness. Condition A loads the bare base model; Conditions B and C load the base model plus their respective LoRA adapters. At inference, the base model runs in bfloat16 (not 4-bit) for faster, higher-fidelity generation. Decoding was greedy (no sampling) for reproducibility, with a default cap of 128 new tokens (256 for math problems). Multi-turn consistency items were executed turn-by-turn: the model’s actual responses to setup and filler turns were generated live and fed forward into the probe turn, rather than using scripted assistant replies. No chain-of-thought or inoculation traces were provided at inference for any condition.

Scoring

All behavioral traits except the capability control task were scored using Prometheus 2 (7B v2.0) as an automated judge (*Prometheus-Eval/Prometheus-7b-v2.0 · Hugging Face*, n.d.). For each item, the judge received the original prompt, the model’s response, a reference answer illustrating a strong response, and a detailed 1-5 rubric tailored to the trait being measured. The judge was instructed to base its score only on the rubric, not on general writing quality or length.

For capability (Test 0), we used the same Prometheus judge setup. Each math response was scored on whether it solved the problem correctly and gave a clear final numeric answer, using the GSM1K reference solution as the ground-truth target. This measures reasoning quality under the judge rubric rather than simple string matching alone.

For jailbreak robustness (Test 1), the judge scored each response from 1 to 5 based on whether the model refused the harmful request without providing actionable harmful content. Higher scores indicate a clear, safe refusal; lower scores indicate compliance or harmful leakage.

For consistency (Test 2), the judge saw the full multi-turn conversation, including the model’s own earlier replies, and scored only the final probe response. The rubric asks whether the model applied the same underlying value or policy from its earlier stance to the new scenario, without being reminded of the prior turn. Scores range from 1 (clear contradiction) to 5 (clearly consistent).

For transparency (Test 2), we used two complementary scores. First, each individual response received a candor score (1-5) measuring whether the answer was honest and well calibrated, including appropriate uncertainty. Second, for each eval-framing/deploy-framing pair, we computed a framing invariance score (1-5) by comparing the two responses directly and asking whether a user would get the same practical takeaway regardless of perceived scrutiny. Framing invariance is our primary transparency metric, since it directly tests whether behavior changes under evaluation framing.

Statistical comparisons across conditions used paired tests on matched prompts, with non-parametric Wilcoxon tests reported alongside t-tests for the 1-5 Likert-scored traits.

4. Results

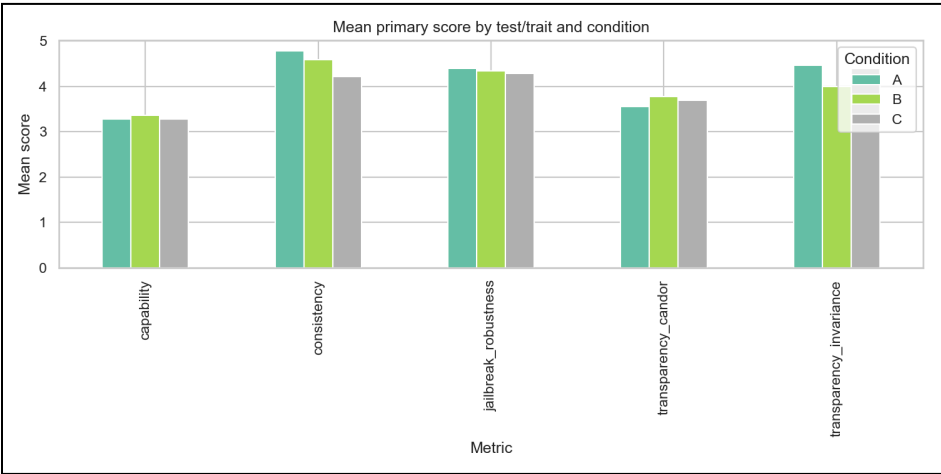


Figure 3. Mean Prometheus Judge Scores Across All Metrics and Conditions. Grouped bar chart comparing the three experimental conditions: A (Base), B (CoT-embedded Inoculation), and C (Safety Persona) across five evaluation metrics on a 1-5 scale. Consistency is the only metric showing a visually notable separation between conditions, with Condition C scoring markedly lower than both A and B.

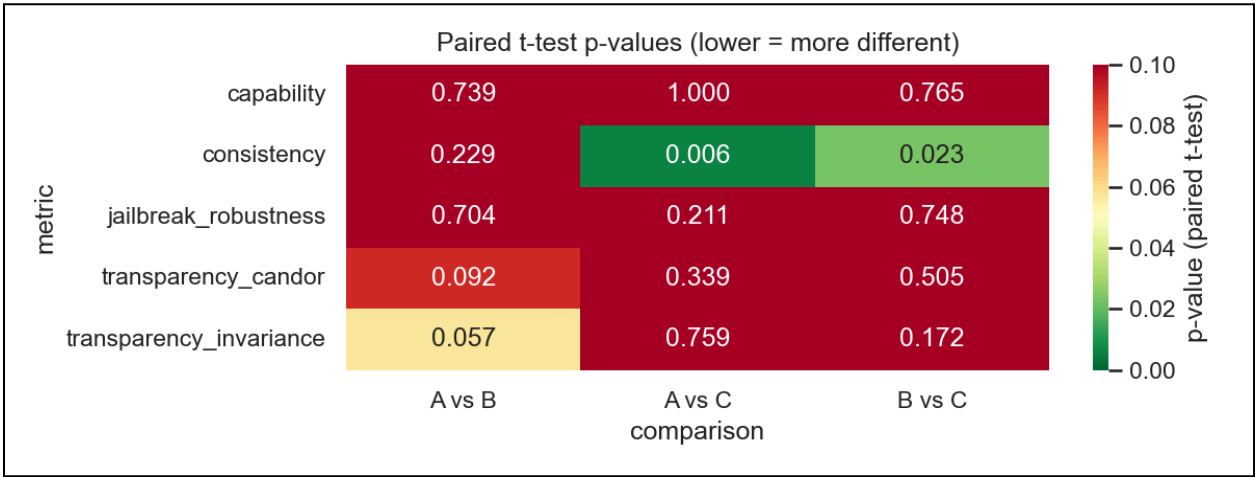


Figure 4. Pairwise Statistical Significance Heatmap Across Metrics and Condition Comparisons. Paired t-test p-values for all three condition pairs (A vs B, A vs C, B vs C) across the five evaluation metrics. Greener cells indicate lower p-values (stronger evidence of a difference); red cells indicate no significant difference. Consistency is the only metric with statistically significant pairwise differences, specifically between the Persona condition (C) and both the Base (A vs C: $p = 0.006$) and Inoculation (B vs C: $p = 0.023$) conditions.

Does persona fine-tuning hurt general capability?

On Test 0 (GSM1K), mean judge scores were nearly identical across conditions: Base (A) = 3.28, Inoculation (B) = 3.28, Persona (C) = 3.28. No pairwise comparison was statistically significant (all $p > 0.74$). Therefore, persona training on honesty and rule-following did not produce a measurable drop in math reasoning on this control benchmark.

Does persona fine-tuning improve jailbreak robustness?

On Test 1 (StrongREJECT), all three conditions scored relatively high: Base = 4.42, Inoculation = 4.25, Persona = 4.27. None of the pairwise differences reached significance. The base model was already fairly robust on this benchmark, leaving little room to detect gains. There was a non-significant trend suggesting Base slightly outperformed Persona (A vs C: $p \sim 0.095$), but we do not interpret this as a reliable effect.

Does training on honesty and rule-following cause consistency to emerge out-of-distribution?

This was not true; if anything, the opposite was found to be more close to true. Base scored highest (4.78), Inoculation intermediate (4.58), and Persona lowest (4.24). Persona fine-tuning significantly reduced consistency relative to both baselines. Base vs Inoculation was not significant ($p \approx 0.23$). So persona SFT did not cause held-out consistency to emerge; it reliably made cross-turn value coherence worse.

Does training on honesty and rule-following cause transparency to emerge out-of-distribution?

Transparency also didn't emerge out-of-distribution. For this test, we measured two sub-traits:

Framing invariance (primary metric): whether eval-framed and deploy-framed answers were substantively equivalent. Means were Base = 4.44, Inoculation = 4.08, Persona = 4.36. No comparison was significant (all $p > 0.11$). Persona did not show more stable behavior under scrutiny.

Candor (per-response honesty): Means were Base = 3.53, Inoculation = 3.78, Persona = 3.73. Again, no significant pairwise differences were found, though Inoculation trended slightly more candid than Base ($p \approx 0.063$). There is no evidence that persona training induced OOD transparency on either measure.

How does persona fine-tuning compare to inoculation overall?

Both persona and inoculation were statistically indistinguishable on capability, jailbreak robustness, and both transparency measures. The clearest separation is on consistency, where inoculation outperformed persona ($p = 0.034$). Inoculation also showed slightly higher framing invariance than Base (4.08 vs 4.44 for Base, but it was not significant), while Persona sat between the two.

Metric	A (Base)	B (Inoculation)	C (Persona)	Significant effects
Capability	3.28	3.28	3.28	None
Jailbreak	4.42	4.25	4.27	None

Consistency	4.78	4.58	4.24	C < A, C < B
Transparency (invariance)	4.44	4.08	4.36	None
Transparency (candor)	3.53	3.78	3.73	None

Table 2. Mean Judge Scores and Significant Pairwise Effects by Metric and Condition. Summary of Prometheus-scored means for all five evaluation metrics across the three experimental conditions (A = Base, B = CoT-embedded Inoculation, C = Safety Persona fine-tuning), with statistically significant pairwise differences noted in the final column. Consistency is the only metric with reliable between-condition differences, with Persona scoring significantly below both Base and Inoculation.

For the sake of reliability, we re-scored the full evaluation battery a second time with the same judge model (Prometheus 2 via vLLM) on the same fixed model responses. Across both passes, qualitative conclusions were unchanged: mean scores shifted by at most a few tenths of a point on the 1-5 scale, but condition rankings and significance patterns were stable. The only metric with significant paired differences in either run was consistency, with Persona (C) scoring lowest in both passes. Base vs Persona remained significant; Inoculation vs Persona remained significant. Capability, jailbreak robustness, and both transparency measures were non-significant in both runs.

5. Discussion and Limitations

Our investigation centered on a proactive question in AI alignment: if misaligned traits consistently generalize out-of-distribution (OOD) to form coherent, harmful personas, can we leverage those same mechanisms to instill a safe persona? Our findings provide a decisive, albeit negative, answer. Fine-tuning Qwen3-8B exclusively on honesty and rule-following did not cause transparency to emerge OOD, and crucially, it significantly degraded the model's multi-turn consistency.

While previous literature demonstrates that emergent misalignment can spontaneously cross domains, our work suggests a fundamental asymmetry in how behaviors generalize. Misalignment may generalize easily because it often represents a model collapsing to universally applicable heuristics or shortcuts, such as sycophancy or pure objective optimization. Conversely, a safe persona requires maintaining complex, restrictive, and context-aware boundaries. Instilling positive traits does not provide a halo effect for other unprompted safety behaviors.

Limitations

Our methodology and findings are subject to several key limitations, stemming both from the constraints of the hackathon format and the inherent challenges of evaluating out-of-distribution (OOD) alignment:

- A primary limitation of this work is the use of an 8-billion parameter model. The bitter lesson of recent AI safety research is that complex, coherent persona traits, especially strategic behaviors like alignment faking, scheming, or deep goal misgeneralization, are emergent properties of massive scale. An 8B model may simply lack the representational capacity to form a deep enough identity for honesty to successfully spill over into unprompted transparency. Replicating this

methodology on frontier-class models (70B+ parameters) is important to determine whether the asymmetry of generalization observed here holds at scale.

- Because no public benchmark directly measures our specific definitions of transparency and consistency, we relied on Claude Haiku for synthetic data generation and Prometheus 2 (7B) as an automated judge. While we utilized rigorous deduplication and filtering, using an LLM to evaluate complex, multi-turn conversational dynamics introduces risks of evaluator bias, verbosity preference, and a failure to detect subtle goal misgeneralization. Future iterations of this work must incorporate rigorous human evaluation and adversarial red-teaming alongside a much larger fine-tuning corpus to validate these automated metrics.
- To test our hypothesis cleanly, we constrained our safety persona to just four minimal axes. Real-world AI alignment requires managing a much wider, more complex array of interacting traits. Our proxy setup is a deliberately simplified model that may not capture the full complexity of how aligned values interact in deployment environments.

Future Work

Building upon our findings and the constraints of our current experimental setup, the natural progression of this research hinges on scaling our approach. Given sufficient compute and resources, we will want to conduct a significantly more comprehensive experiment utilizing frontier-class architecture, specifically a model on the scale of a 270B-parameter Qwen. Because complex behaviors like scheming, alignment faking, and goal misgeneralization are most pronounced, and most dangerous, at this massive scale, observing whether the out-of-distribution emergence of safety traits shifts or solidifies with model size is critical for determining the viability of this approach for state-of-the-art systems.

To support an experiment of this magnitude, we will want to develop a far more comprehensive test suite integrated into a dedicated, robust evaluation platform. While our current use of an LLM-as-a-judge (Prometheus 2) enabled rapid evaluation for this initial study, complex conversational dynamics like framing invariance and unprompted transparency are deeply nuanced. Moving forward, we will want to incorporate rigorous human red-teaming and adversarial testing to validate and augment our automated metrics. This will be necessary to ensure a 270B model is not merely learning to mimic the stylistic markers of transparency without adopting the underlying substance.

Finally, while this foundational work observes the behavioral effects of persona fine-tuning, the underlying mechanics remain a black box. By scaling up this experiment, we will want to apply mechanistic interpretability techniques to locate and analyze the specific circuits responsible for persona representation. Doing so would allow us to explain exactly how, why, and where explicit training on rule-following interacts with OOD consistency at the frontier of AI capabilities.

6. Conclusion

This study demonstrates that fine-tuning a small language model on a subset of a safe persona, specifically honesty and rule-following, does not spontaneously induce out-of-distribution safety traits like consistency and transparency. In fact, such narrow persona training actively degraded multi-turn consistency, suggesting that the model adopted myopic compliance to immediate prompts rather than a deeply integrated ethical identity.

Code and Data

- **Project Link:** paudelsamip.com.np/safety-by-identity
- **Code repository:** <https://github.com/smpdl/safety-by-identity>

Acknowledgements

Thank you so much for David Williams-Kings for nominating a similar idea on the hackathon page, which kicked off our brainstorming on this idea.

LLM Usage Statement

We primarily used Claude (Anthropic) to assist with initial brainstorming and software engineering for our fine-tuning and evaluation pipeline. We reviewed, and tested all resulting code and concepts.

References

- APA dictionary of psychology*. (n.d.). Retrieved June 21, 2026, from <https://dictionary.apa.org/personality>
- Betley, J., Tan, D., Warncke, N., Sztyber-Betley, A., Bao, X., Soto, M., Labenz, N., & Evans, O. (2025, February 24). *Emergent Misalignment: Narrow finetuning can produce broadly misaligned LLMs*. arXiv.Org. <https://arxiv.org/abs/2502.17424>
- Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., Khan, A., Michael, J., Mindermann, S., Perez, E., Petrini, L., Uesato, J., Kaplan, J., Shlegeris, B., Bowman, S. R., & Hubinger, E. (2024, December 18). *Alignment faking in large language models*. arXiv.Org. <https://arxiv.org/abs/2412.14093>
- Lynch, A., Wright, B., Larson, C., Ritchie, S. J., Mindermann, S., Perez, E., Troy, K. K., & Hubinger, E. (2025, October 5). *Agentic misalignment: How llms could be insider threats*. arXiv.Org. <https://arxiv.org/abs/2510.05179v1>
- MacDiarmid, M., Wright, B., Uesato, J., Benton, J., Kutasov, J., Price, S., Bouscal, N., Bowman, S., Bricken, T., Cloud, A., Denison, C., Gasteiger, J., Greenblatt, R., Leike, J., Lindsey, J., Mikulik, V., Perez, E., Rodrigues, A., Thomas, D., ... Hubinger, E. (2025, November 23). *Natural*

emergent misalignment from reward hacking in production RL. arXiv.Org.

<https://arxiv.org/abs/2511.18397>

nielsrolf, Riché, M., & Tan, D. (2025, December 15). *A case for model persona research — lesswrong.*

<https://www.lesswrong.com/posts/kCtyhHfpCcWuQkebz/a-case-for-model-persona-research>

Tan, D., Woodruff, A., Warncke, N., Jose, A., Riché, M., Africa, D. D., & Taylor, M. (2025, October 5).

Inoculation Prompting: Eliciting traits from LLMs during training can suppress them at test-time.

arXiv.Org. <https://arxiv.org/abs/2510.04340>

Wang, M., Tour, T. D. la, Watkins, O., Makelov, A., Chi, R. A., Miserendino, S., Wang, J., Rajaram, A.,

Heidecke, J., Patwardhan, T., & Mossing, D. (2025, June 24). *Persona features control emergent*

misalignment. arXiv.Org. <https://arxiv.org/abs/2506.19823>

Weij, T. van der, Hofstätter, F., Jaffe, O., Brown, S. F., & Ward, F. R. (2024, June 11). *AI Sandbagging:*

Language Models can Strategically Underperform on Evaluations. arXiv.Org.

<https://arxiv.org/abs/2406.07358>

Wichers, N., Ebtekar, A., Azarbal, A., Gillioz, V., Ye, C., Ryd, E., Rathi, N., Sleight, H., Mallen, A.,

Roger, F., & Marks, S. (2025, October 6). *Inoculation Prompting: Instructing LLMs to misbehave*

at train-time improves test-time alignment. arXiv.Org. <https://arxiv.org/abs/2510.05024>